ENR 112: LINEAR ALGEBRA LABORATORY

PROJECT REPORT

Name	Prina Patel
Enrolment No.	AU2340040

PROJECT TITLE: Finding Rate Of Chemical Reaction Using Matrix Algebra

1. Introduction-

In this project, we focus on modelling and simulating the concentrations of reactants andproducts that are involved in a coupled chemical reaction. The reaction will be as follows-

$$A \rightarrow B \rightarrow C$$

A –*-k1-* **–> B–** *-k2-* **–> C**

Here, A is the reactant, B is the intermediate, and C is the product. k1 and k2 are the rateconstants that determine the rate at which each substance is consumed or formed.

We will use matrix algebra (matrix exponentiation) to solve the system of ordinary differentialequations (ODEs) that controls the concentrations of the substances that are involved in coupled chemical reactions.

2. Objective-

The main objective of this project is to

- Simulate the concentrations of the chemical substances A, B, and C over time.
- Using matrix algebra to solve the system of differential equations.
- Visualizing the concentrations of the substances over time through plots.

3. Modelling the system (Equations)-

Substance A is converted to B, so the concentration of A will decreaseTherefore,

the equation for the substance A will be:

$$\frac{d[A]}{dt} = -k1[A]$$

Substance B is intermediate. B first increases when A is converted to B, then decreases when Bis further converted to C.

Therefore, the equation for substance B will be:

$$\frac{d[B]}{dt} = k1[A] - k2[B]$$

Substance C is the product. C initially starts at 0 and increases when B is converted into C. Itcontinues to rise as the time passes.

Therefore, the equation for substance C will be:

$$\frac{d[C]}{dt} = k2[B]$$

1. [A], [B], and [C] are the concentrations of substances A, B, and C, respectively.

2. k1 and k2 are rate constants governing the speed of the conversion of A to B and B to C, respectively.

Now, the system of equations can be written in the matrix form as follows:d (

[A][B][C]) / dt =[−k1, 0 , 0 ; k1, −k2, 0 ; 0, k2, 0] * ([A][B][C])

$$\frac{d}{dt}\begin{matrix}[A]\\[B]\\[C]\end{matrix} = \left|\begin{matrix}-k1 & 0 & 0\\ k1 & -k2 & 0\\ 0 & k2 & 0\end{matrix}\right| * \begin{matrix}[A]\\[B]\\[C]\end{matrix}$$

Let K = rate constant matrix

$$K = \left|\begin{matrix}-k1 & 0 & 0\\ k1 & -k2 & 0\\ 0 & k2 & 0\end{matrix}\right|$$

Let C = concentration vector

$$C = \begin{bmatrix} [A] \\ [B] \\ [C] \end{bmatrix}$$

The solution to this system can be represented as:

$$C(t) = \exp(Kt) * C0$$

Where C0 is the initial concentration vector, and exp (Kt) is the matrix exponential of the rateconstant matrix multiplied by time.

4. Procedure – For MATLAB

4.1- Define Rate Constant Matrix

The matrix K can be defined as follows:

$$K = \begin{vmatrix} -k1 & 0 & 0 \\ k1 & -k2 & 0 \\ 0 & k2 & 0 \end{vmatrix}$$

Let k1=0.5 and k2=0.3 be the rate constants for the reaction.

4.2- Define Initial Concentrations

Let the initial concentrations of the substances be:

$$C0 = \begin{bmatrix} [1] \\ [0] \\ [0] \end{bmatrix}$$

So, initially, only substance A will be present at concentration 1, while B and C will haveconcentrations of 0.

4.3- Time Range

Here, let the time range be from 0 to 20 with 100-time steps.

4.4- Using Matrix Algebra to Solve ODEs

Matrix algebra can be used to calculate the concentration of A, B, and C.

$$C(t) = \exp(Kt) * C0$$

To compute the matrix exponential of Kt, the **expm** function of MATLAB is used. Multiplying it bythe initial concentration vector gives the concentrations at that time point.

4.5- Plotting Results

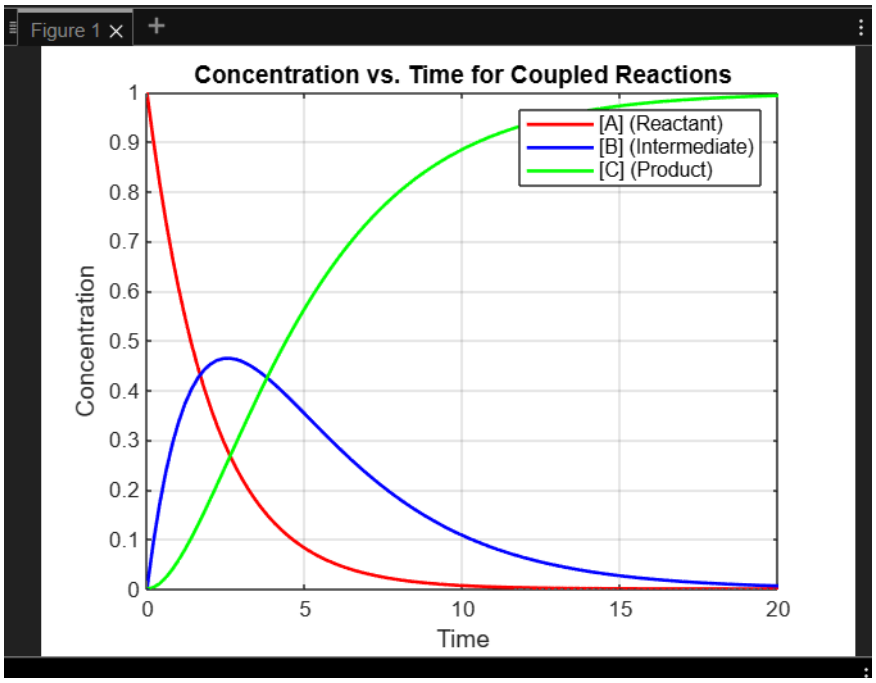
The concentrations of A, B, and C are plotted against time using MATLAB's plot function.

MATLAB CODE-

```
% PROJECT- Calculating Rate Of Chemical Reaction Using Matrix Algebra

% Step 1: Defining the rate constant matrix

%let rate constant for A to B = k1k1 =

0.5

%let rate constant for B to C = k2k2 =

0.3


K = [

   -k1, 0, 0;

    k1, -k2, 0;

    0, k2, 0

]


% Step 2: Defining the initial concentrations

% for substance A, B, C respectivelyC0 =

[1; 0; 0]


% Step 3: Defining the time range (0 to 20) with 100 steps

t = linspace(0, 20, 100);
```

```
% Step 4: Solving for concentrations using matrix algebraC =
zeros(3, length(t));

%C stores concentrations of ([A], [B], [C])

%number of rows are 3. It corresponds to the three
% substances ([A], [B], and [C])

%the number of columns are (length(t))
%It corresponds to number of time points.

for i = 1:length(t)
   C(:, i) = expm(K * t(i)) * C0; % Matrix exponential solutionend

% Step 5: Plotting the concentrations over timefigure;
plot(t, C(1, :), 'r-', 'LineWidth', 1.5, 'DisplayName', '[A] (Reactant)'); hold on;
plot(t, C(2, :), 'b-', 'LineWidth', 1.5, 'DisplayName', '[B] (Intermediate)');
plot(t, C(3, :), 'g-', 'LineWidth', 1.5, 'DisplayName', '[C] (Product)');hold off;

% Step 6: Adding labels and legend
xlabel('Time');
ylabel('Concentration');
title('Concentration vs. Time for Coupled Reactions');
legend('show', 'Location', 'northeast');  %show legend with labels from 'DisplayName'grid on;
```

5. Result-



Graph from MATLAB

(Representation of the changes in concentrations of reactants and products according to therate constants)

The concentrations of A, B, and C are plotted with respect to time

1. From the graph, we can see that A gradually decreases from 1 to 0. A is converted to B,so its concentration decreases.
2. In the graph, **B** first increases (as A is converted to B), then it decreases (as B is furtherconverted to C).
3. **C** initially starts at 0 and then increases when B is converted into C. It continues to riseas time passes.

6. Applications:

- Chemical Engineering
- Medicine and Biochemistry
- Nuclear Chemistry
- Energy Production
- Food Industry